VISUAL INTERVIEW & LEARNING CHEAT SHEET · POSTER 1

Statistical Foundations

Statistics estimates unknown quantities, quantifies uncertainty, distinguishes signal from noise, and supports decisions when truth is not directly observed.

START WITH THE ESTIMAND

QUESTION
What do we want?

→

ESTIMAND
What quantity?

→

ANALYSIS
What evidence?

Question → Estimand → Analysis

COMMON ESTIMANDS

Population mean · proportion · mean difference · treatment effect · correlation · regression coefficient · odds ratio

WHY FIRST?

A test or model is useful only when it targets the quantity the question actually asks about.

CORE PRINCIPLE

Do not start with “Which test?” Start with “What quantity am I estimating?”

Population, parameter & point estimate

Different samples produce different estimates of the same unknown population quantity.

POPULATION

The group you want to understand.

PARAMETER

Unknown population quantity

μ

POINT ESTIMATE

Single-number estimate

$\bar{X}$ estimates μ

SAMPLING VARIABILITY

Sample 1 → $\bar{X}_1$ → **Sample 2** → $\bar{X}_2$ → **Sample 3** → $\bar{X}_3$

Estimates differ because samples differ—even under the same population and procedure.

MEMORY HOOK

Parameter = target.
Estimate = sample-based guess.

Variance, SD & SE

These quantities describe different layers of variation.

THREE MEANINGS

Quantity	What it describes
Variance	Average squared spread around the mean
Standard deviation	Spread of individual observations
Standard error	Sampling variability of an estimate

MEAN STANDARD ERROR

$$SE(\bar{X}) = s / \sqrt{n}$$

n ↑ ⇒ SE ↓ at 1/√n

FOUR-TIMES RULE

Half the SE → about 4× the sample

MEMORY HOOK

SD describes observations.
SE describes estimates.

Confidence vs credible interval

Both express uncertainty, but probability attaches to different objects.

CONFIDENCE INTERVAL

Estimate ± Critical Value × SE

A 95% procedure captures the true parameter in about 95% of repeated samples.

CREDIBLE INTERVAL

Given the model, prior, and observed data, the parameter has 95% posterior probability of lying inside the interval.

AVOID

Not: “There is a 95% probability this specific interval contains the fixed parameter.”

MEMORY HOOK

CI = Long-run Coverage
CrI = Posterior Probability

Null & alternative hypotheses

Hypothesis tests compare observed evidence with a specified null model.

NULL HYPOTHESIS

$$H_0: \mu_A - \mu_B = 0$$

Usually encodes no difference or no effect at a specified null value.

ALTERNATIVE

$$H_A: \mu_A - \mu_B \neq 0$$

DECISION LANGUAGE

Reject H_0

or

Fail to reject H_0

IMPORTANT NUANCE

We do not literally prove or "accept" the null hypothesis.

Type I & Type II error

Testing decisions can create false alarms or missed detections.

TYPE I · A

$$P(\text{Reject } H_0 \mid H_0 \text{ true}) = \alpha$$

False alarm

TYPE II · B

$$P(\text{Fail to reject } H_0 \mid H_A) = \beta$$

Missed signal

SIGNIFICANCE LEVEL

α is the pre-specified Type I error threshold. .05 is common—but convention, not law.

DECISION TRADE-OFF

Lowering α reduces false alarms but can increase missed detections unless sample size or signal increases.

MEMORY HOOK

Type I = false alarm.
Type II = missed signal.

The p-value

A p-value is a probability about data under the null—not a probability that the null is true.

DEFINITION

Assuming the null hypothesis and statistical model are true, the probability of a result at least as extreme as observed.

$$p = P(\text{Data this extreme or more} \mid H_0)$$

EXAMPLE · P = .03

If the null model were true, results at least this extreme would occur about 3% of the time under repeated sampling.

NOT THIS

× $P(H_0 \mid \text{Data})$
× “probability the result occurred by chance”

MEMORY HOOK

$$p = P(\text{Data} \mid H_0)$$
$$\text{not } P(H_0 \mid \text{Data})$$

Sample size & p-values

Large samples increase sensitivity to small departures from the null.

WHAT LARGE N DOES

Makes small effects detectable.

It does not make those effects important.

WHAT TO INSPECT

Effect Size + CI + Decision Threshold

not only $p < .05$

INTERPRETATION

Statistical detectability and practical importance become increasingly distinct as precision grows.

MEMORY HOOK

Large n makes small effects detectable –not important.

Effect size & practical significance

Evidence, magnitude, and decision relevance are different questions.

EXAMPLES

Mean difference: $\bar{X}_T - \bar{X}_C$

Standardized d = $(\bar{X}_T - \bar{X}_C)/s$

Risk difference: P(Y=1|T)−P(Y=1|C)

Correlation: r

DECISION INGREDIENTS

Magnitude + Uncertainty + Cost + Risk + Guardrails

DECISION LENS

Compare the estimated effect and CI with the minimum practically meaningful effect.

MEMORY HOOK

Significance = evidence.
Effect size = magnitude.
Practical significance = decision relevance.

Power

Power is the probability of detecting a specified alternative under a specified design.

DEFINITION

Power = 1 − β

Probability of rejecting H_0 when the assumed alternative effect is truly present.

POWER RISES WHEN

n ↑
Effect size ↑
Noise ↓
α ↑

DIRECTION

A justified one-sided test can have greater power in its specified direction than a two-sided test.

80% CONVENTION

80% is common—not universal. Required power depends on missed-effect consequences, collection cost, decision importance, and design.

MEMORY HOOK

Power is conditional on the assumed effect and study design.

Sample size & MDE

Plan around the inferential or decision objective—not around the sample already available.

ESTIMATION GOAL

Confidence level · margin of error · outcome variability · expected proportion

TESTING GOAL

Alpha · desired power · effect size / MDE · variance / base rate · allocation · design

MINIMUM DETECTABLE EFFECT

Smallest effect the planned study has adequate power to detect.

KEY PRINCIPLE

Choose the meaningful effect first, then determine the sample needed—not the other way around.

One-sided vs two-sided tests

Direction is part of the design and scientific question.

TWO-SIDED

$$H_A: \theta \neq 0$$

Use when meaningful outcomes could occur in either direction.

ONE-SIDED

$$H_A: \theta > 0 \quad \text{or} \quad \theta < 0$$

Use only when one direction is scientifically or decision-relevant.

TIMING RULE

Choose before seeing results.

Do not switch because the two-sided p-value nearly reached .05.

MEMORY HOOK

Direction is a design choice–not a rescue strategy.

Multiple testing

Testing many hypotheses increases the opportunity for false discoveries.

WHERE IT APPEARS

Many metrics · outcomes · subgroups · feature comparisons

BONFERRONI

$$\alpha^* = \alpha / m$$

Strong family-wise error control; can be conservative.

HOLM

Sequential FWER correction

Often preferable to simple Bonferroni when applicable.

BENJAMINI-HOCHBERG

$$p_{(i)} \leq (i/m)q$$

Controls false discovery rate; useful for larger exploratory programs.

KEY PRINCIPLE

Confirmatory → stronger FWER control.
Exploratory → FDR may fit.

Which basic test?

Test choice follows the estimand and dependence structure.

QUESTION → COMMON METHOD

Question	Common method
One mean vs benchmark	One-sample t-test
Two independent means	Two-sample / Welch t-test
Same people twice	Paired t-test
Two categorical variables, sparse	Fisher exact
Paired binary outcome	McNemar
>2 independent means	ANOVA
Continuous association	Correlation / regression

SELECTION SEQUENCE

Estimand → **Outcome type** → **Dependence** → **Assumptions** → **Method**

T-test vs Z-test

The “use Z when n ≥ 30” shortcut is not a reliable rule.

Z FRAMEWORK

Classically used when the relevant population variance is known, or when a justified asymptotic procedure uses a normal reference distribution.

T FRAMEWORK

Used when population variance is estimated from the sample—the common case for mean comparisons.

CONVERGENCE

df ↑ ⇒ t → Normal

The practical difference shrinks as degrees of freedom increase.

INTERVIEW RULE

If population variance is unknown, use the t framework rather than switching automatically to Z at n=30.

Distribution shape & skew

Mean–median–mode patterns are useful intuition, not universal mathematical laws.

SYMMETRIC UNIMODAL

Mean ≈ Median ≈ Mode

RIGHT SKEW

Mode < Median < Mean

LEFT SKEW

Mean < Median < Mode

WHY SKEW MATTERS

Usefulness of mean · small-sample SEs · sensitivity to extremes · model choice

BOUNDARY

A skewed raw variable does not automatically invalidate regression or require a nonparametric test.

Mean vs median

The right summary depends on the quantity relevant to the decision.

MEAN

Uses every observation.

Sensitive to extreme values; useful for aggregate burden or totals per unit.

MEDIAN

Middle observation.

More resistant to extremes; useful for a typical experience.

RESPONSE-TIME EXAMPLE

Median: typical experience
Mean: aggregate burden
Tail percentile: worst experiences

MEMORY HOOK

Summary statistic follows the decision.

Correlation

Association is not identification of a causal pathway.

RANGE

$$-1 \le r \le 1$$

Strength and direction of association on a specified scale.

CAUSAL BOUNDARY

Correlation ≠ Causation

Association can reflect causality, reverse causality, confounding, selection, or chance.

PEARSON

Linear association

SPEARMAN

Rank-based monotonic association

Useful for monotonic nonlinear patterns or substantively meaningful ranks.

CORRECTION

Do not choose Spearman only because a variable looks non-normal.

Missing data

Mechanism matters more than an arbitrary percentage threshold.

MECHANISMS

Type	Meaning
MCAR	Missingness unrelated to observed or unobserved values
MAR	Explained by observed information
MNAR	Still depends on unobserved values after conditioning

POSSIBLE APPROACHES

Complete case · weighting · multiple imputation · model-based methods · sensitivity analysis

MULTIPLE IMPUTATION

Creates plausible completed datasets, analyzes each, and combines estimates while preserving uncertainty.

AVOID UNIVERSAL RULES

There is no valid “delete below 30%, impute above 30%” rule.

IMPORTANT CORRECTIONS

Mean/median replacement can understate uncertainty and distort relationships. A missing category may help prediction but does not solve the mechanism.

MEMORY HOOK

Missingness mechanism matters more than missingness percentage.

Outliers & influential observations

An unusual observation may be wrong, rare-but-real, or influential—those are different issues.

ASK FIRST

Wrong? Rare-but-real? Influential?

DIAGNOSTICS

Boxplots · scatterplots · standardized residuals · leverage · Cook's distance

NOT VIF

VIF evaluates multicollinearity—not outliers.

INVESTIGATE

Data quality · substantive plausibility · influence on conclusions

RULE

Do not automatically delete extreme values.

Multicollinearity: know the idea

Overlapping predictor information hurts separation of effects more than overall prediction.

CONSEQUENCES

Unstable coefficients · larger SEs · sign changes · difficulty separating effects

DIAGNOSTIC

$$VIF_j = 1 / (1 - R_j^2)$$

No universal cutoff automatically invalidates a model.

POSSIBLE RESPONSES

Remove / combine redundancy · redefine by question · regularize for prediction · dimension reduction

MEMORY HOOK

Multicollinearity hurts separation of effects more than prediction.

Parametric vs nonparametric

Choose the method based on the estimand, structure, assumptions, and robustness—not a desired p-value.

PARAMETRIC

Specified model structure

t-test · ANOVA · linear regression

RANK-BASED ALTERNATIVES

Mann–Whitney · Wilcoxon signed-rank · Kruskal–Wallis

DECISION QUESTIONS

What estimand? What dependence? Which assumptions matter? How robust is the conclusion?

AVOID

Do not switch methods because the p-value is “almost significant.”

KEY PRINCIPLE

Method choice precedes result inspection.

Frequentist vs Bayesian uncertainty

The frameworks answer uncertainty questions using different probability statements.

FREQUENTIST

Parameters are fixed; repeated-sampling behavior determines uncertainty.

SEs · confidence intervals · p-values

BAYESIAN

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

$$p(\theta|\text{Data}) \propto p(\text{Data}|\theta)p(\theta)$$

DIRECT PROBABILITY

$$P(\theta > \delta \mid \text{Data}) = .93$$

δ can represent a meaningful decision threshold.

INTERPRETATION CONTRAST

Frequentist: procedure behavior.
Bayesian: posterior uncertainty given model, prior, and data.

Decisions under uncertainty

Statistical evidence is one input to a product or policy decision.

DECISION EQUATION

Effect + Uncertainty + Cost + Risk + Guardrails + Reversibility

EXAMPLE THRESHOLD

Roll out only if Effect > 2

Ask which effects remain plausible and whether they cross the practical threshold.

NOT ENOUGH

“Is p < .05?”

Statistical significance does not encode costs, risks, or action constraints.

KEY PRINCIPLE

Statistical significance thresholds and decision thresholds are different.

Common interview traps → better answers

Replace the shortcut with the correct conditioning statement, uncertainty principle, or diagnostic.

NINE TRAPS

× "p=.03 means a 3% chance the result happened by chance."
Under the null model, results this extreme or more would occur about 3% of the time.

× "Large samples make p-values meaningless."
They can make tiny effects detectable, so magnitude and practical importance matter more.

× "Not significant means no effect."
The estimate may be too uncertain to distinguish from zero.

× "80% power means an 80% chance this study is significant."
Power is conditional on an assumed true effect and study design.

× "If n≥30, use a Z-test."
With unknown population variance, the t framework remains appropriate.

× "A 95% CI gives 95% probability the parameter is inside."
That is a credible-interval interpretation, not the standard frequentist CI interpretation.

× "Delete cases if less than 30% are missing."
Missingness mechanism matters more than an arbitrary percentage.

× "VIF detects outliers."
VIF diagnoses multicollinearity; residual, leverage, and influence diagnostics address unusual cases.

× "More data solves statistical problems."
More data reduces sampling uncertainty but does not automatically fix bias, confounding, measurement error, or design.

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Statistical reasoning flow

Move from a clearly defined target to an uncertainty-aware decision.

TEN-STEP FLOW

1 · Population + estimand

2 · Study design

3 · Effect summary

4 · Point estimate + SE + interval

5 · Hypotheses + α set first

6 · Effect magnitude

7 · Practical threshold / MDE

8 · Multiplicity

9 · Assumptions + missingness + outliers

10 · Decision: cost, risk, guardrails

EVIDENCE BUNDLE

Point Estimate + SE + Interval

MEANING BUNDLE

Magnitude + Practical Threshold

DECISION BUNDLE

Cost + Risk + Guardrails + Reversibility

CLOSING PRINCIPLE

Translate evidence into the decision—do not substitute a significance rule for judgment.

30-second interview answer

Lead with the estimand, quantify uncertainty, interpret evidence correctly, and connect the plausible effect range to action.

READY-TO-USE ANSWER

"I'd start by defining the estimand rather than jumping straight to a statistical test. I'd report the point estimate together with its uncertainty and focus first on the effect size and confidence interval. If I use a p-value, I'd interpret it as how unusual the observed data are under the null model, not as the probability the null is true. I'd also consider power, multiplicity, missing-data assumptions, and whether the study design supports the inference. Most importantly, I'd compare the plausible effect range with the minimum practically meaningful effect and combine magnitude and uncertainty with costs, risks, and guardrails rather than using statistical significance alone as the decision rule."

ANSWER ARCHITECTURE

Estimand → Estimate + interval → Correct p-value → Power + multiplicity → Missingness + design → Practical decision

STRONG CLOSING

Magnitude and uncertainty inform the decision; significance alone does not make it.

Statistical foundations recall board

Use five contrasts to reconstruct the full interview response under pressure.

ESTIMATION

Point Estimate + Uncertainty

TESTING

H_0 → p → Evidence

ERRORS

Type I = False Alarm

Type II = Missed Signal

INTERVALS

CI = Coverage

CrI = Posterior Probability

DECISION

Effect + Uncertainty + Cost + Risk

MOST IMPORTANT RULE

Ask how large the effect is, how uncertain it is, and whether it matters–not only whether p<.05.